

The Deadly Race

A Formal Threat Model for Physical-Coercion (“Wrench”) Attacks,
and a Construction that Clears It

Great Wall coercion-resistance paper series

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Abstract

Physical-coercion (“wrench”) attacks defeat custody schemes that key management and cryptography treat as out of scope. We give a *formal model* of coercion against a secret-holder and prove that a large class of “graceful” recovery designs — those that leave a device seizure holding a *feasible-but-slower* path to the secret — are strictly worse than useless: they manufacture a material incentive to *assassinate* the victim. We name this the **Deadly Race**, derive it from a single impossibility (one cannot prove one has forgotten), and distil a design bar — the **No-Gray-Area Principle** — that any coercion-resistant scheme must clear. We then present a construction, **Great Wall**, that clears the bar for the Bitcoin self-custody setting while keeping custody *individual*, and we locate a shipping alternative — timelock-rescue vaults — precisely: it clears the same bar by the *other* available route, at the cost of a trusted third party. The framework fixes a bar; it does not claim Great Wall is the unique construction that clears it.

1 Introduction

Coercion is the unmodeled attack. Standard threat models — Dolev–Yao on the wire [4], IND-CCA for encryption, forward secrecy for sessions — all assume the adversary cannot compel the *holder* of a secret. For bearer assets that assumption is empirically false: the physical attack registry maintained by Lopp [9], and the systematic study of the phenomenon in the financial-technologies literature [1], document a growing series of incidents in which a Bitcoin holder is physically threatened with a “\$5 wrench” until they surrender their keys.

Cryptographic hardness does not help here. A wrench attacker does not break the cipher; they compel the person who can operate it. Worse, the standard defence offered to holders — *obscurity* (“never tell anyone you own Bitcoin”) — is not a security mechanism at all in the sense of Kerckhoffs [8]: it is a bet that the attacker lacks knowledge the defender has, and it degrades precisely as adoption spreads that knowledge.

This paper models coercion directly. Our contributions are:

1. a **formal coercion model** (§2) with an explicit adversary, a Kerckhoffs premise, and an engineering “feasibility line” that turns asymptotic security into concrete parameter bounds;
2. the **Deadly-Race impossibility chain** (§3): a classical impossibility (you cannot prove you forgot) that escalates, step by step, into an incentive to murder the victim;
3. the **No-Gray-Area Principle** (§3), the design bar that a coercion-resistant scheme must clear, together with a *reachability* dichotomy that classifies every clearance route;
4. a **construction** (§4), the Great Wall orbit, that meets the bar with individual custody, including a proof that the master secret is at least 64* from every seizable state; and
5. a **taxonomy** (§5) that classifies existing designs by *which* clearance route they take — illustrated on shipping timelock-rescue vaults.

For citation, we name the central results after their first statement in the Great Wall design record [6]: the **Deadly-Race Lemma** (*Villas Boas’ Lemma*) and the **No-Gray-Area Principle** (*Villas Boas’ Principle*).

2 Model

2.1 Principals and capabilities

Definition 1 (Principals). The **holder** $\mathcal{V}$ possesses (in memory and on devices) whatever is needed to reach a master secret K that gates an asset. The **coercer** $\mathcal{A}$ seeks K . Optional **heirs** or **delegates** may hold partial capability on $\mathcal{V}$'s behalf.

Assumption 1 (Kerckhoffs, premise 0). $\mathcal{A}$ knows the protocol in full: all algorithms, all source, $\mathcal{V}$'s setup topology, and every in-scope parameter. Only the protocol *secrets* are hidden. Every result below is proved *against* such an $\mathcal{A}$; security by obscurity is assumed to contribute nothing.

Assumption 2 (Adversary power). $\mathcal{A}$ is Dolev–Yao on the wire [4]: it observes all transmitted plaintext and ciphertext but cannot decrypt beyond its computational budget. $\mathcal{A}$ may *seize any device* and read all state on it (vault blobs, time-lock blobs, commitment scripts, review schedules). $\mathcal{A}$ has *unbounded compute within a wall-clock custody window*, and may *physically coerce* $\mathcal{V}$ for a bounded duration — bounds calibrated against the incident durations in the registry [9].

2.2 The feasibility line

Asymptotic “hard” and “easy” are useless against a fixed, resourced attacker; we replace them with two engineering stipulations that fix what *feasible* means and turn the lemmas into concrete bounds.

Assumption 3 (Feasibility stipulations). Let a *cheap* evaluation be one call to a non-memory-hard hash H , and a *memory-hard* evaluation one call to an hours-scale Argon2d-class function H^* [2].

- **S1 (feasible)**. 2^{32} cheap evaluations are within $\mathcal{A}$'s budget.
- **S2 (infeasible)**. 2^{64} memory-hard evaluations — written **64*** — are beyond it.

Two currencies span the gap between them. *Entropy* buys $32 \rightarrow 64$: a second independent 32-bit stage. But plain 2^{64} *cheap* work still sits at the border of a resourced attacker's reach (hours), so the $64 \rightarrow 64^*$ step must be bought by *memory-hardness* — which is why the gating function has to be Argon2-class. We refer to plain cheap 2^{64} throughout as the *softest number*: it must be named explicitly wherever it is invoked, never silently treated as infeasible.

2.3 The tacit-knowledge premise

The construction of §4 rests the secret on *tacit* knowledge. We state the premise abstractly here; §4 instantiates it.

Assumption 4 (Tacit knowledge, TKBA). The secret is a set of fractal locations that are *recognisable but not transmissible*.

- **TKBA₁**. A point is intransmissible without the fractal parameters that render it; $\mathcal{V}$ cannot dictate it in the abstract.
- **TKBA₂**. The *deep* fractal parameters are not a hand-overable artifact; producing them requires walking a memory-hard chain that later locations do not exist before.

3 The Deadly-Race impossibility chain

The core results form an escalating chain: each follows from the one before, starting from a classical impossibility and ending in a design bar.

Lemma 1 (Non-Negative-Proof Lemma, NNPL). *No agent can furnish satisfactorily verifiable proof of lack of access to a given piece of information — whether that access would be through their own memory, or directly or indirectly through physical media. One can prove one knows something; one cannot prove one has forgotten it, nor that no backup of it exists anywhere.*

Argument. Knowledge admits a positive certificate: exhibit the datum, or answer a challenge that only a knower could answer. Non-knowledge admits no such certificate. Any protocol in which $\mathcal{V}$ demonstrates ignorance is one in which a knowing $\mathcal{V}$ could simulate the same transcript by withholding what they know; and no observation of $\mathcal{V}$ or of any finite set of media excludes the existence of a further, unobserved copy. The asymmetry is information-theoretic, not a matter of effort. $\square$

Lemma 2 (Deadly-Race Lemma / Villas Boas’ Lemma). *Because a released $\mathcal{V}$ cannot prove (Lemma 1) that they retain no resumable copy of their protocol state, any seizure that leaves $\mathcal{A}$ with enough information to eventually reach K — however much slower than legitimate use — defines a **race** between $\mathcal{A}$ (working from the seized information) and the released $\mathcal{V}$ (presumed to hold complete information). Whoever reaches K first seizes whatever it gates. The existence of that race creates a **material incentive for $\mathcal{A}$ to assassinate $\mathcal{V}$** — and possibly the heirs — upon seizing sufficient information, so as to eliminate the competing racer.*

Proof. Suppose the seizure leaves $\mathcal{A}$ on a feasible path to K of wall-clock length T_A , and $\mathcal{V}$, once released, on a path of length $T_V \leq T_A$. By Lemma 1, $\mathcal{A}$ cannot verify any promise by $\mathcal{V}$ to abstain or to have destroyed their copy, so $\mathcal{A}$ must assume $\mathcal{V}$ races. Eliminating $\mathcal{V}$ deletes the competing racer and converts an uncertain win into a certain one; it is therefore a strictly dominant action for a payoff-maximising $\mathcal{A}$ whenever the marginal cost of the killing is below the contested value. The incentive is a structural consequence of the race, independent of $\mathcal{A}$ ’s temperament. $\square$

Principle 1 (No-Gray-Area Principle / Villas Boas’ Principle). *A coercion-resistant protocol must, for a fixed threat model, admit **only** outcomes in which a wrench attack clearly succeeds or clearly fails — never a race. A protocol in which some wrench-attack outcome is a race is, by construction, not acceptable: the gray area is precisely where the assassination incentive of Lemma 2 lives.*

Operational restatement. Seizing a device must yield either *nothing feasible* (clear failure) or *the secret itself while $\mathcal{A}$ is present and $\mathcal{V}$ is captive* (clear success, no free racer) — never a feasible-but-slower offline path that $\mathcal{A}$ could pursue after releasing or killing $\mathcal{V}$. “Feasible” is judged against the attacker of Assumptions 2–3, not against a casual one. $\square$

Remark 1 (Reachability — the remote-racer escape). The assassination incentive of Lemma 2 is realizable *only against a racer $\mathcal{A}$ can reach*. A racer beyond physical reach — e.g. a remote rescuer or delegate — cannot be eliminated, so their racing advantage stands and the attack fails cleanly. Remoteness of the racer is therefore itself a remediation. Hence there are exactly **two honest ways to clear Principle 1**:

1. **remove the race** — admit no feasible seizure path at all (custody stays individual); or
 2. **remove $\mathcal{A}$ ’s reach over the racer** — interpose a remote delegate (custody becomes shared).
- This dichotomy is the classification axis of §5. The escape buys safety by moving the winning racer *off the individual* — it requires a trusted third party.

4 Construction — Great Wall (clears by removing the race)

We now exhibit a construction that clears Principle 1 by the first route of Remark 1: it admits *no feasible seizure path*, so custody remains individual. Full normative detail is in the canonical spec [5]; we give the parts the theorem needs.

4.1 The orbit

Let H be a cheap, entropy-preserving one-way hash and H^* an memory-hard (Argon2-class) hash of tunable cost. Let σ be a public, precomputation-resistant *timechain* salt whose cross-user amortization is negligible (the companion Namtso primitive [3]). Define an **orbit** of N deep stages:

$$o_0 = H(\sigma), \tag{1}$$

$$o_i = H^*(H(o_{i-1} \parallel Sh_{i-1})), \quad 1 \leq i \leq N, \tag{2}$$

where the raw inputs $\{o_{i-1}, Sh_{i-1}\}$ are **wiped** in the instant between the cheap inner hash and the long memory-hard step. Computing $c := H(o_{i-1} \parallel Sh_{i-1})$ first (instant), wiping, then running the long $o_i := H^*(c)$, means the raw prior orbit point and Shamir secret exist only momentarily — not across the whole memory-hard window. This directly shrinks the window in which K is seizable, and is the reason not to write $H^*(o_{i-1} \parallel Sh_{i-1})$ directly.

4.2 No single-point oracle

Each stage i carries not an individual point but a *whole Shamir polynomial* [12]. Stage i has s_i points $p_{i,j}$, each a 32-bit leaf-area of a fractal $\theta_{i,j} = H(o_i \parallel j)$; any t_i of them reconstruct the degree- (t_i-1) polynomial Sh_i of $t_i \cdot 32 \geq 64$ bits ($t_i \geq 2$). The orbit absorbs the *full* polynomial Sh_i , not its constant term $f(0)$.

Remark 2 (Block-orbit $k \geq 2$ condition). Because the smallest object any orbit link commits is a full stage (≥ 64 bits), no individual point $p_{i,j}$ is ever checkable in isolation — there is no sub-threshold oracle against which a single guessed point could be verified. This is the block-orbit $k \geq 2$ condition, realised as Shamir $t_i \geq 2$ per stage.

4.3 Tacit multi-fractal stages

The points $p_{i,j}$ instantiate Assumption 4: they are recognisable on their fractal boards but not transmissible without $\theta_{i,j}$ (TKBA₁), and the deep $\theta_{i,j}$ are regenerated from o_i — which itself requires walking the memory-hard chain (TKBA₂). Using several shallow-zoom fractals per stage lets a stage carry $t_i \cdot 32$ bits at low zoom depth per board, keeping each point recognisable while tuning stage entropy in 32-bit steps.

4.4 Cheap terminal secret

The master secret is

$$K = H(o_N \parallel Sh_N), \tag{3}$$

deliberately using the *cheap* H . A memory-hard last step would only *prolong* the window in which o_N and the final points are live (paying an H^* *after* the last point is recognised), worsening the very window we shrink in (2). The resistance therefore lives in *entropy*: standard deep stages carry 96 bits ($t_i = 3$), so a cheap ledger brute of Sh_N from a bare o_N costs 2^{96} — infeasible — while the terminal window stays instant.

4.5 The 64* floor

Theorem 1 (64* floor on the master secret). *Under Assumptions 2–4 and the standard build of (2)–(3), from every seizable state the master secret K is at least 64* away: no single seizure leaves K fewer than two memory-hard 32-bit stages from reach. Consequently a device seizure yields nothing feasible — a clear failure in the sense of Principle 1, with no racer for $\mathcal{A}$ to reach.*

Proof by case analysis over seizable states. From a seized o_i ($i \leq N$), reaching K requires guessing $Sh_i, \dots, Sh_N$; each Sh_j guess is $2^{t_j \cdot 32}$ and is H^* -gated, since it is checked only by recomputing the memory-hard link (2). Preimage resistance of H, H^* makes the orbit forward-only: a seized o_i cannot yield o_{i-1} . We enumerate:

- *Pure device seizure* (no coercion; only time-lock-gated [11] blobs at rest). All tacit points are required; K is unreachable.
- *Wrench of stage 0* ($p_{0,}$, always present since σ is public). $\mathcal{A}$ computes o_1 with one H^* ; from o_1 , K costs $2^{\sum_{j \geq 1} t_j \cdot 32}$, which is 2^{96} or more in the standard ($N \geq 2$) build — infeasible.
- *Mid-orbit seizure* at o_i ($i \leq N$). K costs $\geq 2^{96}$, each guess H^* -gated — infeasible.
- *Residual o_N* present. Treated separately in §4.6.

Every case except the residual leaves $K \geq 64^*$ — two memory-hard stages, beyond S2 of Assumption 3. $\square$

Remark 3 (Why a last-stage KDF fails). The floor is not automatic. In a naive sequential setup whose master secret is $KDF(p_N)$ — determined by the *last* stage alone — a surprise seizure at stage $N-1$ holds the preimage of stage N ; deriving it and brute-forcing the final 32-bit point costs 2^{32} (S1, feasible). That is a feasible-but-slower path: a gray area, rejected by Principle 1. The requirement is that K be $\geq 64^*$ from *every* seizable state, which is exactly the no-single-point-oracle / block-orbit ($k \geq 2$) condition met above.

4.6 The tolerated residual window

One gray sliver survives: a wrench attack that lands with o_N imminently or currently live can, after the *second-to-last* point of stage N is surrendered, leave $\mathcal{A}$ one cheap $2^{32} \cdot H$ brute ((3) is cheap) — roughly a one-minute window — from K . We report this honestly as a bounded residual rather than claim zero. It is tolerated on five grounds:

1. a well-trained fractal is recognised in ~ 1 minute;
2. $\mathcal{A}$'s plan already budgets time-lock or re-derivation, subdue, and interface time — a residual minute rarely changes the escape;
3. assassinating *trusts* that the earlier surrendered points were genuine; one wrong point and the coercion retry is burned;
4. it sharply *worsens judicial liability* — murder qualified by futile motive *and* by the victim's defenselessness, atop armed robbery; and
5. the terminal K is deliberately cheap (§3), so keeping memory-hardness in the orbit rather than the terminal step removes any advantage $\mathcal{A}$ would gain by pre-staging.

Whether the window opens after the (t_N-1) -th or (t_N-2) -th point rests on the softest number of Assumption 3 — inside the tolerated reasoning either way. Net: a tiny window in which $\mathcal{A}$ is, more likely than not, incentivised *not* to act.

5 Classification of existing designs

Remark 1 gives a classification axis: a design that clears Principle 1 does so either by *removing the race* or by *removing $\mathcal{A}$'s reach over the racer*. The framework should classify a shipping design, and it does — via the reachability remark, not by calling it broken.

5.1 Timelock-rescue vaults

Take **Rewind Bitcoin** [10] as the exemplar. Funds sit in a Bitcoin-native timelock vault; unlocking starts a multi-day countdown; if an unauthorized unlock is triggered, the owner *or a trusted delegate* has that window to press “Rescue” and sweep to a cold panic address.

Against *remote* theft it is sound: a thief who steals keys trips the countdown, the owner notices, and rescues. But the spend capability sits behind *only a delay*, with keys on the seized

device and no recall floor — so a wrench attacker who seizes the device *does* hold a feasible path (unlock, wait out the countdown). That is a race, in the sense of Lemma 2. Rewind’s answer is exactly the second route of Remark 1: it puts the winning racer out of $\mathcal{A}$ ’s reach — a *remote delegate* the attacker cannot find or eliminate presses Rescue — so the attack fails cleanly. This is a legitimate clearance, but the remote-racer kind: its cost is that **custody is no longer individual**.

Remark 4 (When the escape closes). The remedy holds only while the delegate is genuinely remote. If the delegate is a friend who happens to be visiting during the attack, *both* problems apply at once — custody was already non-individual, and the racer is now co-located and reachable — so the design becomes deadly-race-susceptible again. The user is thus forced into an awkward corner: trust someone they can rarely be near, and accept a perverse incentive to avoid physical proximity to a trusted friend.

5.2 Great Wall, and the honest comparison

Great Wall clears by the *other* route: the secret is tacit and never on the device (Assumption 4, Theorem 1), so a seizure yields nothing feasible — there is no racer to reach, and the time-lock is only a delay *on top of* an already-clean failure. Custody stays individual. The two designs therefore clear the same bar by the two different routes of Remark 1:

Design	Clearance route	Cost / consequence
Timelock-rescue (Rewind)	Remove reach over racer	Remote delegate $\Rightarrow$ <i>shared</i> custody; escape closes if delegate co-located
Great Wall (orbit)	Remove the race	Tacit secret, $\geq 64^*$ floor $\Rightarrow$ custody stays <i>individual</i>

“Put the racer out of reach” versus “leave no race to run” is the honest comparison. Other designs slot onto the same axis: plausible-deniability and duress-PIN schemes fail Assumption 1 (an educated $\mathcal{A}$ persists past the decoy); Shamir-only social recovery moves reach onto guardians; multisig-by-geography attempts a weak form of the second route.

6 Quantum note

A few-minutes memory-hard H^* per element also defeats Grover [7]: a quantum search over an n -bit space needs $\sim 2^{n/2}$ *coherent* evaluations of the memory-hard oracle, which is impractical for an hours-scale Argon2d barrier. Hence sub-256-bit standards (96 or 128 bits per stage) are quantum-defensible *without* a 256-bit seed — the memory-hardness that buys the 64^* floor classically also caps the quantum speed-up.

7 Limitations and scope

Out of scope for this model: adversarial-agent scenarios (testator/heir mutual distrust, malicious interference with rotation); device side-channels (timing, EM, acoustic) during active tacit recall; long-horizon cryptographic erosion (century-plus hardness of the underlying primitives and of Argon2 parameters against future hardware); and mass on-chain correlation beyond the privacy of standard taproot outputs and private channels.

Two scoping notes bear repeating. First, the framework fixes a *bar* (Principle 1); it does not claim Great Wall is the unique construction that clears it. Second, the residual window of §4.6 is a genuine, bounded gray sliver, argued to be incentive-safe rather than eliminated — we do not over-claim it away.

8 Conclusion

The Deadly Race turns “graceful degradation” from a virtue into a liability whenever the degraded state is a feasible-but-slower path to the secret: by Lemma 2, that state is a race, and the race pays an assassination incentive. The No-Gray-Area Principle is the resulting design bar, and the reachability remark shows there are exactly two honest ways to clear it. The Great Wall orbit clears it by removing the race entirely — an 64^* floor from every seizable state (Theorem 1) — and so keeps custody individual, where timelock-rescue vaults clear the same bar by removing the attacker’s reach at the cost of a trusted third party.

Open items for the full draft. Fully game-based definitions of NNPL, the Lemma, and the Principle with an explicit adversary game and a reduction for the 64^* floor; a quantitative H^* per-evaluation calibration tying S1/S2 to hardware; a precise game-based definition of “coercion-resistance” that the Principle implies; and an empirical grounding of the residual-window argument against the wrench-attack registry [9].

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